

Strong edge colouring conjecture

OPG-46475

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

A strong edge-colouring of a graph G is a edge-colouring in which every colour class is an induced matching; that is, any two vertices belonging to distinct edges with the same colour are not adjacent. The strong chromatic index $s\chi'(G)$ is the minimum number of colours in a strong edge-colouring of G .

Conjecture

$$s\chi'(G) \leq \frac{5\Delta^2}{4}, \text{ if } \Delta \text{ is even,}$$
$$s\chi'(G) \leq \frac{5\Delta^2 - 2\Delta + 1}{4}, \text{ if } \Delta \text{ is odd.}$$